Name: ..... Due: **Next Class / Friday**

C3 The Bass Guitar HW 3 – Music Man StingRay, Active Electronics & RHCP 'Give It Away'

Artist: Red Hot Chili Peppers (Blood Sugar Sex Magik, 1991, Bass: Flea, Producer: Rick Rubin)
https://www.youtube.com/watch?v=Mr_uHJPUlO8

1. "Give It Away" features Flea's aggressive, cutting bass tone recorded direct and through vintage tube amplification. Identify the elements below. (3)

Equipment / Production Element	Hardware Implementation & Engineering Function
Music Man StingRay Bass	Large Alnico humbucker at bridge "sweet spot" providing massive low end and
Active 9V Onboard Preamp	Battery-powered buffer converting high-Z to low-Z; active Baxandall circuit boo
Ampeg SVT 8x10 Cabinet	
Simultaneous DI + Mic Tracking	

2. Explain why the Music Man StingRay's active 9V preamplifier prevents high-frequency signal degradation across long stage or studio cables compared to traditional passive bass volume circuits. (3)

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3. Why did Ampeg engineer the SVT 8x10 cabinet with eight sealed 10-inch drivers in four compartments rather than two large 15-inch woofers? Detail cone mass and acoustic transient response. (3)

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4. According to Mersenne's Law, explain why extended scale lengths (35 inches) or fanned-fret multi-scale designs are commonly used on 5-string basses tuned down to Low-B (30.9 Hz). (3)

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5. Multi-Track Phase Alignment: Explain why simultaneous tracking with a DI box and a miked speaker cabinet introduces acoustic latency delay, and how DAW engineers correct the resulting comb filtering. (4)

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6. Passive Bass Circuit vs Active 9V Bass Circuit Architecture: complete the comparative analysis below: (4)

Passive Bass Wiring (e.g. Vintage Fender P-Bass / J-Bass) (2)
High-Z output, cable capacitance treble rolloff, cut-only RC tone pot bleeding highs to ground, organic dynamic response:
Active 9V Onboard Electronics (e.g. Music Man StingRay / Warwick) (2)
Low-Z buffered output, zero cable length treble loss, active boost and cut Baxandall EQ, high transient headroom:

Total:	**/18**